

CSM-322: Information and Coding Theory

Lab - 6

(Odd Semester 2024-25)

October 17, 2024

Q1. Linear Codes

Let G be a generator matrix for the binary linear code C . Write a program to print all the codewords of C .

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

Q2. Linear Codes

Write a program to verify whether the following matrix is a parity check matrix for the linear code $C = \{0000, 1110, 1011, 0101\}$.

$$H = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$